Three boxes to fill

Animath — Stage olympique de Sophia-Antipolis · 2006 · France

---

In the equation

$$x^3 + \square x^2 + \square x + \square = 0,$$

François first replaces one of the boxes by a non-zero integer. Then Rémy replaces another box by an integer, and finally François replaces the last box by an integer. Show that François can play so that the three solutions of the resulting equation are integers.